

Homework 5

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Problem 1

Yes, a linear model appears plausible for this relationship given the plot below.

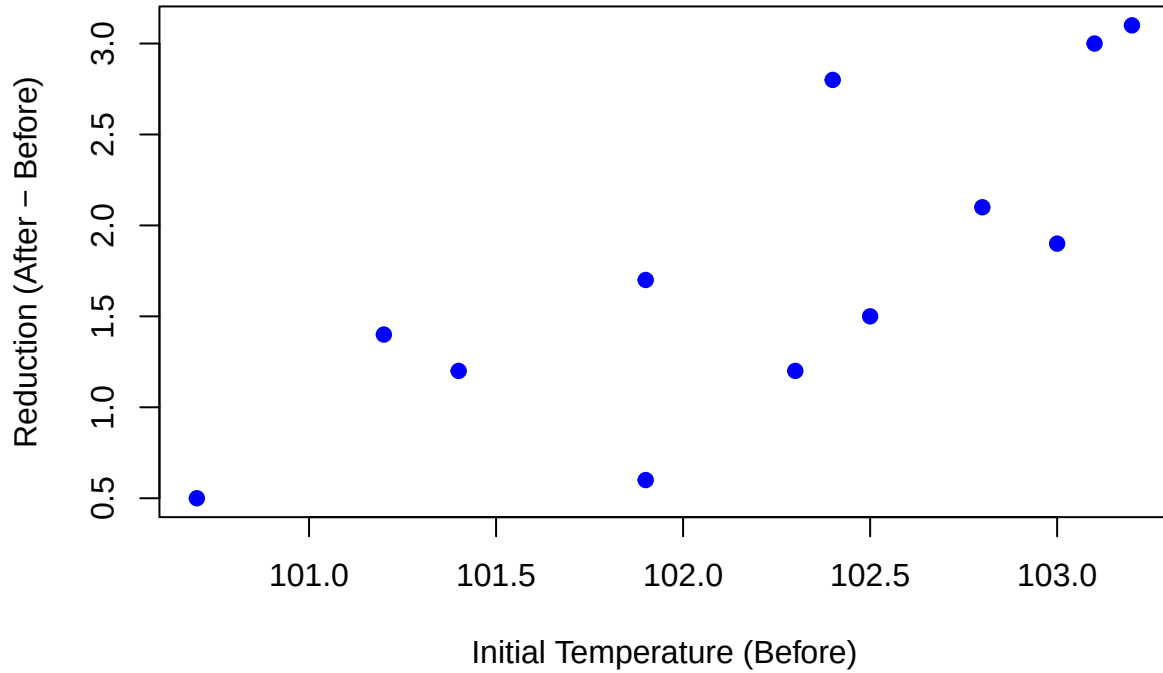
```
patient <- 1:12
before <- c(102.4, 103.2, 101.9, 103.0, 101.2, 100.7, 102.5, 103.1, 102.8, 102.3, 101.9, 101.4)
after <- c(99.6, 100.1, 100.2, 101.1, 99.8, 100.2, 101.0, 100.1, 100.7, 101.1, 101.3, 100.2)

Z <- before - after

df <- data.frame(patient, before, after, reduction = Z)

plot(df$before, df$reduction,
     main = "Temperature Reduction vs. Initial Temperature",
     xlab = "Initial Temperature (Before)",
     ylab = "Reduction (After - Before)",
     pch = 19, col = "blue")
abline(h = 0, lty = 2, col = "gray")
```

Temperature Reduction vs. Initial Temperature



Problem 2

The median value of β_1 is 0.7942. The 95% credible interval is $[0.2991, 1.257]$. This confirms the investigators' suspicion that there is a nonzero relationship between the temperature before taking aspirin and the drop in temperature after taking aspirin for children with influenza. The DIC is 25.281. Dbar is 22.145. pD is 3.136. The posterior density estimate is below.

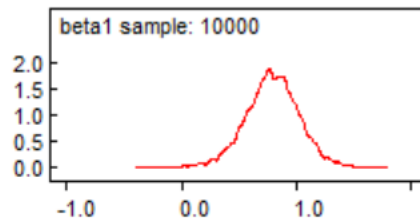


Figure 1: Posterior density of beta1

The model specifications, data, and initial values are as follows:

```
model { for (i in 1:n) { Z[i] ~ dnorm(mu[i], tau) mu[i] <- beta0 + beta1 * X[i] }
beta0 ~ dnorm(0, 0.0001) beta1 ~ dnorm(0, 0.0001) tau ~ dgamma(0.1, 0.1) # Precision has Gamma prior }
list( n=12, X = c(102.4, 103.2, 101.9, 103.0, 101.2, 100.7, 102.5, 103.1, 102.8, 102.3, 101.9, 101.4), Z = c(2.8,
3.1, 1.7, 1.9, 1.4, 0.5, 1.5, 3.0, 2.1, 1.2, 0.6, 1.2) )
list(beta0 = 0, beta1 = 0, tau = 1)
```

Problem 3

The probability that Z_{13} is greater than 0 is calculated below using its samples exported from WinBUGS.

```
samples_df <- read.table("z13_samples.txt", header = FALSE)

z13_samples <- samples_df$V2

prob_z13_gt_0 <- mean(z13_samples > 0)

cat("P(Z[13] > 0) =", prob_z13_gt_0, "\n")
```

```
## P(Z[13] > 0) = 0.4971
```

The posterior density of Z_{13} is shown below. The median is -0.004623.

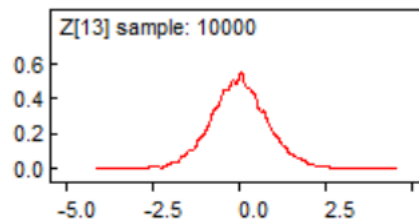


Figure 2: Posterior density of z_{13}

Problem 4

The stats for the standardized residuals are shown below. None of them seem to be outliers as none are above or below 2 standard deviations from the mean.

node	mean	sd	MC error	2.5%	median	97.5%	start	sample
std_res[1]	1.48	0.4373	0.005663	0.649	1.471	2.353	1	10000
std_res[2]	0.9214	0.4944	0.005953	-0.05043	0.9188	1.904	1	10000
std_res[3]	0.3158	0.3181	0.003547	-0.3118	0.3182	0.9387	1	10000
std_res[4]	-0.805	0.4601	0.004987	-1.713	-0.8	0.09411	1	10000
std_res[5]	0.7425	0.5037	0.005529	-0.2543	0.7418	1.715	1	10000
std_res[6]	-0.09008	0.6195	0.006554	-1.31	-0.09536	1.118	1	10000
std_res[7]	-0.8082	0.3651	0.003965	-1.533	-0.8025	-0.1083	1	10000
std_res[8]	0.8876	0.4667	0.005642	-0.0239	0.8869	1.818	1	10000
std_res[9]	-0.2091	0.3683	0.004041	-0.9291	-0.2077	0.512	1	10000
std_res[10]	-1.042	0.3782	0.004167	-1.798	-1.033	-0.3198	1	10000
std_res[11]	-1.509	0.4553	0.005199	-2.432	-1.492	-0.6427	1	10000
std_res[12]	0.1467	0.4145	0.004418	-0.6715	0.147	0.954	1	10000

Figure 3: Stats for standardized residuals

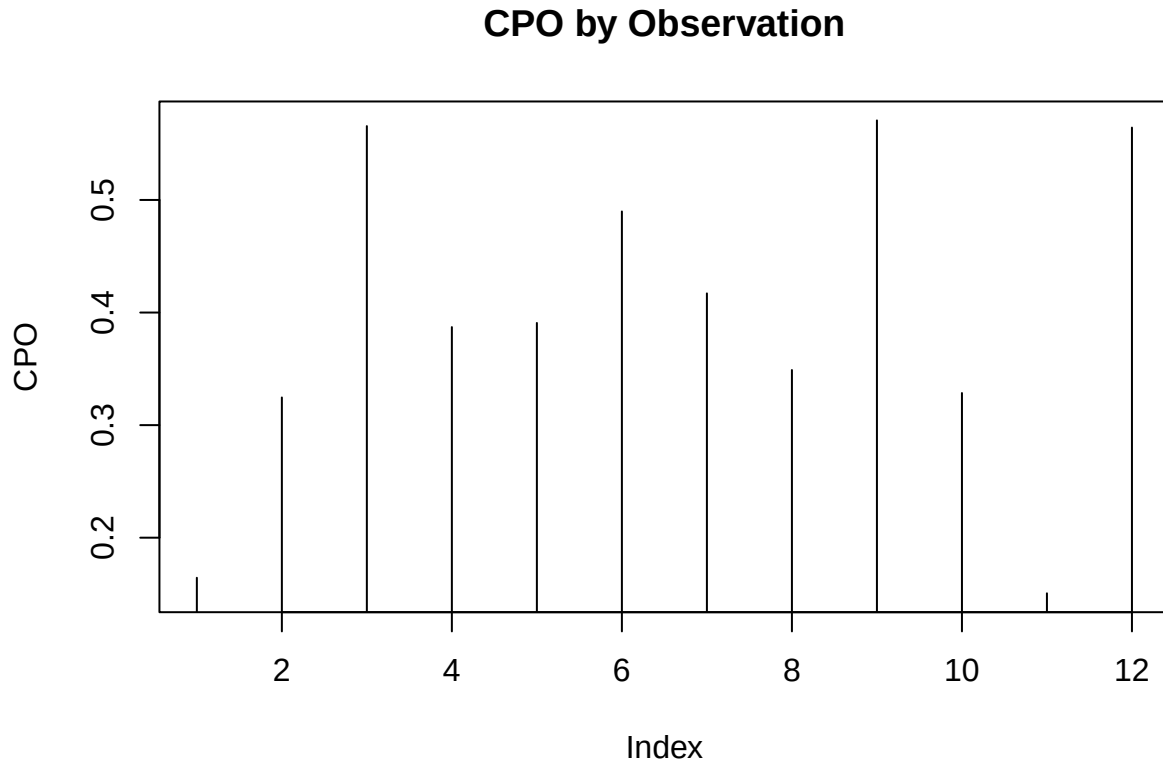
The CPO is calculated below.

```
Z <- c(2.8, 3.1, 1.7, 1.9, 1.4, 0.5, 1.5, 3.0, 2.1, 1.2, 0.6, 1.2)
mu_raw <- read.table("mu_samples.txt", header = FALSE)
tau_raw <- read.table("tau_samples.txt", header = FALSE)
tau_samples <- tau_raw[, 2]
S <- length(tau_samples)
n <- length(Z)
mu_values <- mu_raw[, 2]
mu_matrix <- matrix(mu_values, nrow = S, ncol = n)
lik_matrix <- matrix(NA, nrow = S, ncol = n)
```

```

for (i in 1:n) {
  lik_matrix[, i] <- sqrt(tau_samples / (2 * pi)) * exp(-0.5 * tau_samples * (Z[i] - mu_matrix[, i])^2)
}
cpo <- 1 / colMeans(1 / lik_matrix)
plot(cpo, type = "h", main = "CPO by Observation", ylab = "CPO", xlab = "Index")

```



1st observation:

```
print(cpo[1])
```

```
## [1] 0.1644111
```

11th observation:

```
print(cpo[11])
```

```
## [1] 0.1506217
```

Those are the smallest values, but they do not indicate outlier status.

Problem 5

The new DIC value is 26.803 ($\bar{D} = 22.512$, $pD = 4.291$), which is higher than that of the linear model, suggesting that we should not add the quadratic term.

The new Z_{13} samples have a median of 0.9353. This is very different from the previous median. This may no longer be a valid prediction because $x=100$ is far away from the domain of the rest of the data.